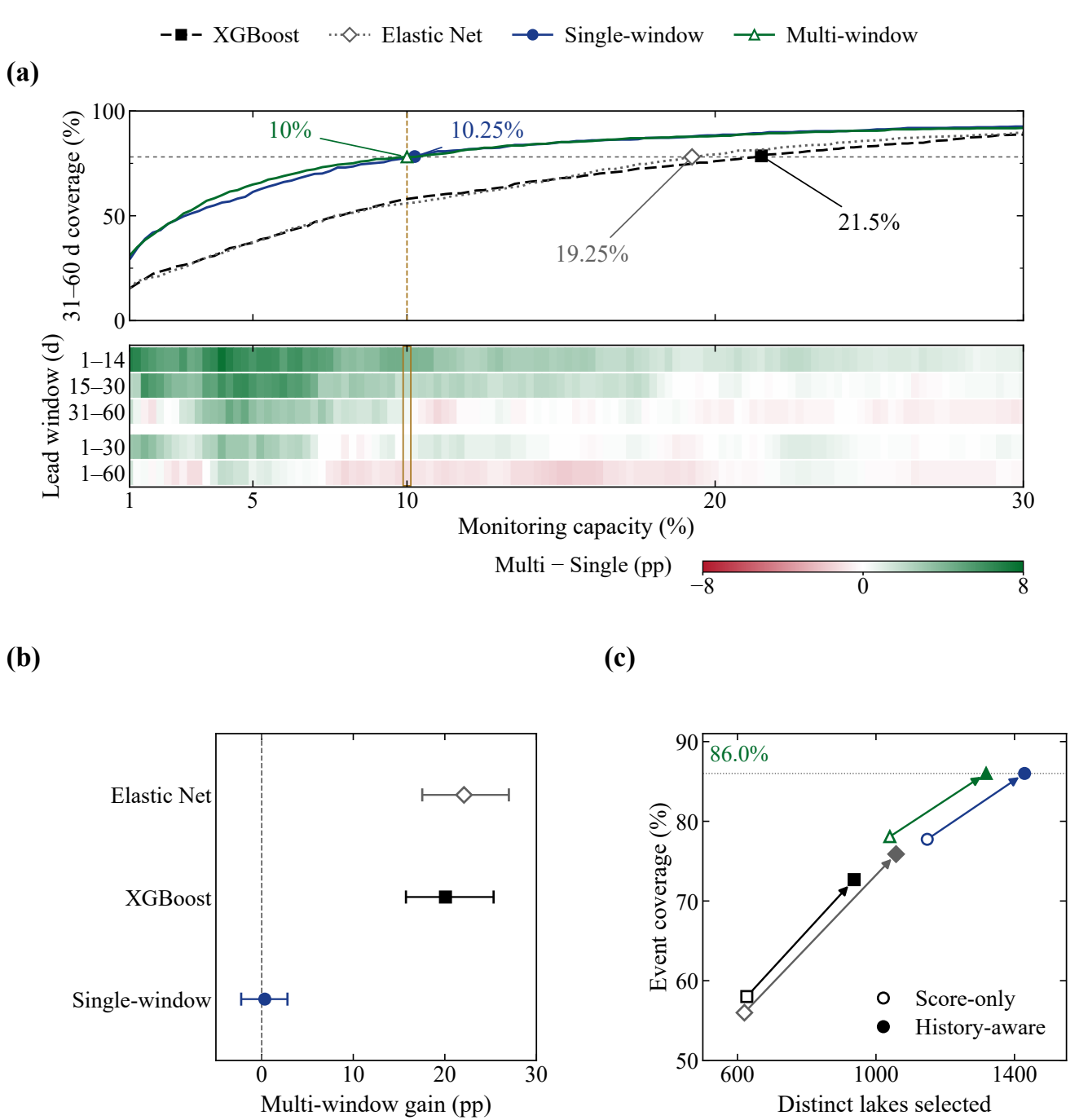

XGBoost
Elastic Net
Single-window
Multi-window
(a)
31–60 d coverage (%)
100
50
0
10%
10.25%
19.25%
21.5%
Lead window (d)
1–14
15–30
31–60
1–30
1–60
1
5
10
20
30
Monitoring capacity (%)
Multi − Single (pp)
−8
0
8
(b)
Elastic Net
XGBoost
Single-window
0
10
20
30
Multi-window gain (pp)
(c)
86.0%
Event coverage (%)
90
80
70
60
50
600
1000
1400
Distinct lakes selected
Score-only
History-aware